

# Form/Function Relativity: The Reference Frame Is the Perspective

B. Wallace (TensorRent)

with computational verification by Claude (Anthropic)

Companion to *Spectral Rigidity and Shuffled Spacing Discriminant* (FF06e) — SIP License v1.1

## Abstract

The shuffle knife (FF06e) sorts each spectral witness into *form* (its value is a property of the distribution class) or *function* (its value reads the object,  $\zeta$ , specifically) by comparing the real zeros against a marginal-matched surrogate. That verdict is stated as if it were a property of the witness. It is not. It is a property of the *pair* (witness, reference measure), and the reference measure — the surrogate ensemble against which the witness is scored — is a free choice. Change the choice and the verdict moves. We make this operational by scoring the same five witnesses against a *ladder* of reference frames ordered by how much structure each preserves: Poisson (no level repulsion), the GUE-marginal shuffle (nearest-neighbour spacing law only), a full-GUE bulk ensemble (the two-point correlation), and  $\zeta$  itself (the arithmetic). Refining the frame turns witnesses from function to form one structural class at a time — a staircase. The repulsion and Wigner-shape witnesses are **function** against Poisson ( $44\text{--}50\sigma$ ) and **form** against the GUE-marginal frame ( $0.1\text{--}0.2\sigma$ ): the identical witness carries opposite labels in the two frames. The rigidity (number-variance) witness is function against both of those and form ( $1.6\sigma$ ) only once the frame preserves the two-point function. The arithmetic prime-resonance witness is function against every frame short of  $\zeta$  itself ( $260\text{--}832\sigma$ ): it is the irreducibly object-specific content. The form/function label is therefore a coordinate on (witness  $\times$  frame), not an intrinsic attribute — *one witness's form is another witness's function*. This is the spectral instance of a general role-relativity in nested systems, the same statement carried at different depths by *one man's trash is another's treasure* and *one party's security is another's leverage*: a component's role is conferred by the enclosing system that resolves it, and a differently-resolving system in the nest confers the opposite role. It is also the same reference measure  $d\mu/d\nu$  that already carries the perspective in the BCH-transfer-entropy lemma of the core monograph; here it is exhibited numerically. Reproducible at seed 20260423. Nothing here proves RH; negatives are kept first-class.

**Publication status (24 September 2026).** This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

## 1 The observation

FF06e defines form and function through a single surrogate ensemble.

**Definition 1** (Form and function, FF06e Def. 2, recalled). A witness  $W$  is a scalar functional of a spectrum. Evaluate it on the real zeros,  $W_{\text{real}}$ , and on an ensemble of marginal-matched surrogates (the unfolded spacings permuted and re-accumulated), with mean  $\overline{W}$  and standard deviation  $\sigma_W$ .

Its *influence* is  $z(W) = |W_{\text{real}} - \overline{W}|/\sigma_W$ ;  $W$  is **form** if  $z(W)$  is small ( $\lesssim 3$ ) and **function** if  $z(W)$  is large.

The definition names one surrogate — the gap-shuffle — and reads off a label. But the gap-shuffle is a choice, and FF06e’s own scope section already records that the choice is load-bearing: the lag-1 verdict swings by  $\sim 6\sigma$  “induced purely by the choice of GUE pipeline,” with two legitimate GUE references disagreeing by more than the candidate signal. If replacing one repulsion-matched reference by another moves the verdict, then replacing a repulsion-matched reference by a *non*-repulsion-matched one (Poisson) will move it much further. The label was never attached to the witness alone. It was attached to the witness *as seen from* a reference frame, and FF06e fixed the frame silently.

We make the frame a variable. Read the other way, Definition 1 is already the folk observation that one man’s trash is another’s treasure, restricted to spectra and made quantitative: the  $z$ -score is the amount by which one enclosing system (the surrogate) fails to resolve what another (the object) contains, and “form” versus “function” is which side of that gap the witness is being read from.

## 2 Relative form and function

**Definition 2** (Form/function relative to a reference frame). Let  $\nu$  be a reference ensemble (a “frame”) on spectra: a distribution from which surrogates are drawn. For a witness  $W$  define

$$z_\nu(W) = \frac{|W_{\text{real}} - \mathbb{E}_\nu[W]|}{\sqrt{\text{Var}_\nu[W]}}.$$

$W$  is **function relative to**  $\nu$  if  $z_\nu(W) > \tau$  (we use  $\tau = 3$ ) and **form relative to**  $\nu$  otherwise. Definition 1 is the special case  $\nu = \text{gap-shuffle}$ .

The verdict is now explicitly a two-argument object. The content of this note is that the second argument matters, and matters in a structured way. Before specialising to spectra, we state the shape of the claim at the level at which it is really operating.

**Principle 1** (Role-relativity in nested systems). *Let a component sit inside a family of systems ordered by nesting: system  $\nu_1$  is nested in  $\nu_2$  (written  $\nu_1 \preceq \nu_2$ ) when every structure  $\nu_1$  resolves is also resolved by  $\nu_2$ . Each enclosing system reads off a two-valued role for the component — the label it assigns given what that system can and cannot resolve. Then the role is a coordinate on (component  $\times$  system), not a property of the component: the same component is assigned opposite roles by different systems in the nest, and ascending the nest can only convert the marked role (the one that records unresolved structure) into the unmarked one, never the reverse. Each component therefore has a single threshold level at which its role flips, once, and stays flipped. One man’s trash is another’s treasure; one party’s security is another’s leverage; one field’s form is another’s function — three readings of a single component by three enclosing systems that resolve it differently.*

The spectral case is the instance of this principle in which the components are witnesses, the role is {form, function}, and the nest is the refinement ladder of §3.

**Principle 2** (Form/function relativity). *The label {form, function} is a coordinate on the product space (witness  $\times$  frame), not a property of the witness. For a fixed witness  $W$  there exist frames  $\nu_1, \nu_2$  with  $W$  form relative to  $\nu_1$  and function relative to  $\nu_2$ ; equivalently, relative to a fixed frame two witnesses split into form and function, and the split is not preserved when the frame changes. One witness’s form is another witness’s function.*

The mechanism is not mysterious. A witness is function relative to  $\nu$  exactly when it reads structure that  $\nu$  does *not* reproduce. So the label tracks the difference between what the witness reads and what the frame preserves — a relation between two things, neither of which is the witness by itself. Order the frames by refinement (how much structure each preserves) and the labels must fall monotonically:

**Proposition 1** (Monotone staircase). *Let  $\nu_1 \preceq \nu_2$  mean that every scalar structure preserved (in ensemble mean) by  $\nu_1$  is also preserved by  $\nu_2$ . If a witness  $W$  reads only structure that  $\nu_2$  preserves, then  $z_{\nu_2}(W) \leq z_{\nu_1}(W)$  up to sampling noise: refining the frame can turn function into form but never the reverse. Hence along a refinement chain each witness has a single threshold frame at which it falls from function to form, and the witnesses are totally ordered by the depth of structure they read.*

This is a statement about the geometry of the (witness  $\times$  frame) table, and it is testable: the table should be a staircase, not a scatter. It is also Principle 1 made a theorem: the staircase is “trash becomes treasure” with the enclosing systems named — a component keeps its marked role until the system around it is large enough to resolve what it reads, then flips exactly once. The relabelling is not a paradox to be resolved but a coordinate to be read off.

### 3 The frame ladder

We instantiate a four-rung refinement chain on the Riemann zeros. Each rung preserves strictly more structure than the last.

| Frame $\nu$     | preserves         | construction                                            |
|-----------------|-------------------|---------------------------------------------------------|
| Poisson         | density only      | i.i.d. exponential spacings, density-matched            |
| GUE-marginal    | + NN-spacing law  | permute the real unfolded spacings, re-accumulate       |
| GUE-full (bulk) | + two-point corr. | eigenvalues of a random Hermitian matrix, bulk unfolded |
| $\zeta$ itself  | + arithmetic      | the real zeros (terminal frame)                         |

The chain is  $\text{Poisson} \preceq \text{GUE-marginal} \preceq \text{GUE-full} \preceq \zeta$ . This is a *nest* of systems in the sense of Principle 1: each frame is an enclosing system that resolves strictly more of the spectrum’s structure than the one before, so each reads a witness’s form/function role differently, and the everyday dualities (trash/treasure, security/leverage) are the same relabelling one nesting level apart. Poisson destroys level repulsion; the GUE-marginal shuffle restores the exact nearest-neighbour spacing distribution but destroys all correlations of order  $\geq 2$  and all arithmetic; the full-GUE bulk restores the RMT two-point function but still carries no arithmetic;  $\zeta$  itself is the only frame that carries the primes, and against it every witness is trivially form ( $z = 0$  against itself). The reference measure  $\nu$  is exactly the free reference  $d\mu/d\nu$  of the BCH–transfer-entropy lemma (core monograph, Lemma 2.9): the information asymmetry, and with it the form/function reading, is always measured against a *chosen*  $\nu$ . Choosing  $\nu$  differently relabels; that choice is what we call the perspective.

We score five witnesses, ordered by the depth of structure they read: *repulsion* (fraction of spacings below half the mean) and *shape* ( $L^2$  distance of the spacing histogram to the GUE Wigner surmise), both reading only the nearest-neighbour spacing law; *lag-1* serial correlation, reading

local order; *rigidity* (number variance in fixed windows), reading the two-point correlation; and *arith* ( $\sum_{p \leq 29} \langle \cos(\gamma \log p) \rangle^2$ ), reading the arithmetic.

## 4 Measurement

On the first  $N = 4000$  Odlyzko zeros at seed 20260423, with 200 surrogate draws for the two exact frames and 40 for GUE-full (bulk, central 80% trimmed before unfolding). Cells give  $z_\nu$  and the resulting label at  $\tau = 3$ .

| witness   | vs Poisson | vs GUE-marginal | vs GUE-full     | reads           |
|-----------|------------|-----------------|-----------------|-----------------|
| repulsion | 49.5 FUNC  | <b>0.2 form</b> | 5.3 FUNC        | NN-spacing law  |
| shape     | 44.3 FUNC  | <b>0.1 form</b> | 6.5 FUNC        | NN-spacing law  |
| lag-1     | 25.7 FUNC  | 25.9 FUNC       | 6.5 FUNC        | local order     |
| rigidity  | 10.7 FUNC  | 10.4 FUNC       | <b>1.6 form</b> | two-point corr. |
| arith     | 259.9 FUNC | 832.2 FUNC      | 479.1 FUNC      | arithmetic      |

Read the rows. **Repulsion and shape** are the theorem in miniature: 44–50 $\sigma$  function against Poisson, 0.1–0.2 $\sigma$  form against the GUE-marginal frame. The identical scalar functional, on the identical zeros, is function in one frame and form in the next — Principle 2 made numerical. Their agreement with the critical line, viewed from Poisson, looks like a sighting of  $\zeta$ ; viewed from a repulsion-matched frame it is revealed as a property of the class that any GUE-spaced sequence would reproduce. **Rigidity** falls one rung later: function against Poisson and against the spacing-marginal shuffle (both fail to reproduce the two-point function), form (1.6 $\sigma$ ) only once the frame is full GUE. **Arith** never falls: it is function against every frame that lacks the primes, and its  $z$  is two to three orders of magnitude above threshold. It is the object-specific content that no non-arithmetic frame can absorb — the explicit-formula coupling, i.e. unique factorisation read spectrally.

The table is a staircase, as Proposition 1 requires: reading down the “threshold frame” at which each witness turns form gives repulsion/shape at GUE-marginal, rigidity at GUE-full, arith beyond every frame short of  $\zeta$ . The witnesses are thereby *ordered* by the depth of structure they read, and that order is an output of the measurement, not an input.

The two exact frames (Poisson, GUE-marginal) run in  $\sim 2$  s and carry the load-bearing flip; the GUE-full column requires dense diagonalisation and is gated behind `--full` ( $\sim 7$  min). The driver is `code/hp_knife_suite/hp_form_function_relativity.py`.

## 5 Why this is the same object as the monograph’s asymmetry

The relativity is not a quirk of the knife; it is the discrete face of a structure already central to the framework.

**The reference measure is the perspective.** The BCH–transfer-entropy lemma writes the information asymmetry as  $\Delta\mathcal{I}(\varepsilon) = \varepsilon \langle f - g, \nabla \log \frac{d\mu}{d\nu} \rangle_\mu + 2\varepsilon^2 \langle [f, g], \nabla \log \frac{d\mu}{d\nu} \rangle_\mu + \dots$ , against a reference measure  $\nu$ . The Radon–Nikodým derivative  $d\mu/d\nu$  is exactly the frame of Definition 2: what counts as asymmetry, and therefore what counts as function, is measured against a chosen  $\nu$ . The knife’s “surrogate ensemble” and the lemma’s “reference measure” are the same slot in the same formula. Form/function relativity is what that slot being free *looks like* on a spectrum.

**Form and Function were already frame-relative in gauge theory.** The monograph assigns Form to the observable curvature and Function to the potential (the vierbein/connection pair, the field-strength/gauge-field pair). That assignment is itself gauge-dependent: which half of  $e^{i\theta} = \cos \theta + i \sin \theta$  is “the observable” depends on the phase origin, and a gauge rotation trades one for the other. The core monograph’s own remark that “the distinction is not merely labelling” is correct *within* a fixed frame and incomplete across frames: within a frame Form constrains Function and vice versa; across frames the assignment rotates. Principle 2 is the number-theoretic instance of that rotation — the same statement the title of the parent programme already makes, that form and function are generated by the asymmetry of a pair and not carried by either member alone.

## 6 What this establishes, and where it stops

A reading note on the ARITH row: its  $z$ -values ( $259.9 \rightarrow 832.2 \rightarrow 479.1$ ) are not monotone along the refinement chain, and Proposition 1 does not require them to be — the monotonicity claim constrains only witnesses reading structure the finer frame *preserves*, so a witness that stays FUNC in every frame may fluctuate in magnitude while its label never changes.

**Establishes.** The form/function verdict of FF06e is frame-relative in the precise sense of Definition 2: on the first 4000 zeros the repulsion and shape witnesses are function against Poisson and form against a repulsion-matched frame, and rigidity is function until the frame preserves the two-point function. The (witness  $\times$  frame) table is a monotone staircase, so the witnesses are totally ordered by the depth of structure they read, with the arithmetic witness strictly deepest. This sharpens FF06e’s single-surrogate categorisation into a frame-indexed one and identifies the frame with the reference measure of the monograph’s asymmetry lemma.

**Does not establish.** Nothing here proves the Riemann Hypothesis, constructs a Hilbert–Pólya operator, or bounds off-line behaviour. “Function relative to  $\nu$ ” means “reads structure absent from  $\nu$ ,” not “proves a property of  $\zeta$ .” That the arithmetic witness stays function against every non-arithmetic frame is consistent with, but not evidence for, RH: it says only that the primes are not reproduced by any GUE-type null, which is the explicit formula, not a theorem about zero locations. The general role-relativity (Principle 1) and the everyday dualities are framing, not measurement: the one instance measured here is the Riemann spectrum (§4, tier T1), and the folk sayings are cited to locate the principle, not as evidence for it.

**Two honest edges.** (i) The GUE-full frame is a finite- $N$ , unfolded ensemble; its column is approximate, and it is the two exact frames that carry the argument. Consistent with that approximation, lag-1 remains nominally function ( $6.5\sigma$ ) against GUE-full where Proposition 1 would place its threshold, and the repulsion and shape cells likewise read  $5.3\sigma$  and  $6.5\sigma$  against GUE-full after having gone form ( $0.2\sigma$ ,  $0.1\sigma$ ) against the exact GUE-marginal frame — all three are finite- $N$  apparatus effects of the same kind FF06e flagged (the  $\sim 6\sigma$  inter-pipeline swing), not claimed beyond-GUE signals and not counterexamples to the once-flipped-stays-flipped staircase, which is stated over the exact frames. (ii) The threshold  $\tau = 3$  is a convention; the staircase structure, not the exact cell labels near the threshold, is the result. Both edges are recorded so the table is not over-read.

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This companion accompanies *Spectral Rigidity and Shuffled Spacing Discriminant* (FF06e) and *Spectral Witness Survival*. All measurements use the real first Riemann zeros and the canonical seed 20260423; the relativity suite is reproducible standalone via `hp_form_function_relativity.py`. SIP License v1.1.

## Amendment: scope and evidence, 24 September 2026

Changing the reference frame or measurement map can exchange which features are treated as input and response. An invariant claim must specify its transformation law and compare like observables. The complete-action study supplies a concrete case: coordinate rescaling preserves generalized mass eigenvalues, whereas scaling a physically different action changes all masses. Those operations must not be conflated. The distinction strengthens the methodology without asserting a universal physical equivalence among domains.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository's **papers/updates/** directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.